

National University of Computer and Emerging Sciences, Lahore Campus



Course: Linear Algebra
Program: BS(EE)
Duration: 180 Minutes
Date: 21st Dec. 2016
Sections: EE 1,2,3
Exam: Final

Code: MT104
Semester: Fall 2016
Marks: 100
Weight: 50%
Page(s): 2
Roll No: 15-4522

Instruction: Programmable calculator is not allowed.

Question # 1: [10] Using Elementary matrices to find the inverse of $A = \begin{bmatrix} 1 & 0 \\ 3 & 7 \end{bmatrix}$. Also verify that $A^{-1} = E_k \dots E_2 E_1 I_2$ for some k . Any other method used to evaluate the inverse will not be considered for marking.

Question # 2: [15] Let $T: R^3 \rightarrow R^3$ be a linear operator defined as $T(X) = AX$, where $A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & 4 & 3 \\ 5 & 5 & -2 \end{bmatrix}$.

- Check whether T is One to One & Onto.
- Find kernel of T & basis for kernel of T .
- Find Range of T & basis for Range of T .
- Find Null Space of T and Row Space of T .
- Show that Null Space is orthogonal complement of Row Space by showing that all vectors are mutually orthogonal to each other using standard Inner Product of vectors.

Question # 3(a): [5+10] Use Equivalent statements to check whether A is diagonalizable?

$$A = \begin{bmatrix} 1 & 7 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

b) Use diagonalization process to determine P , D and P^{-1} for B , where $B = \begin{bmatrix} 7 & -2 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$.

Question # 4: [5+5] Suppose $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$ & $B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$. Define Inner Product on A & B as $\langle A, B \rangle = \text{tr}(B^t A)$ (Where $\text{tr}(C)$ means trace of a matrix C).

- Show that it is a valid Inner Product by checking Inner Product Axioms.
- Also compute $\langle A, B \rangle = \text{tr}(B^t A)$ and $\|A\|$ if $A = \begin{bmatrix} 2 & 5 \\ 0 & -3 \end{bmatrix}$ & $B = \begin{bmatrix} 1 & 4 \\ 7 & -5 \end{bmatrix}$.

Question # 5: [10] Let $T(x_1, x_2, x_3) = (x_1, x_2, 0)$ & $T^*(x_1, x_2, x_3) = (-x_1, x_2, x_3)$ be two Linear Operators from R^3 to R^3 .

- Find matrix of transformation for T & T^* .
- Find $T \circ T^*$ & $T^* \circ T$.
- Under what condition(s) $T \circ T^* = T^* \circ T$?

$$P^{-1} B P =$$

$$\begin{bmatrix} 4 & 2 & 0 \\ -4 & 2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

Question#6: [10+10] Let $T(x_1, x_2) = (x_2, -5x_1 + 13x_2, -7x_1 + 16x_2)$ be a Linear Transformation from R^2 to R^3 defined as $T(X) = AX$. Consider two different set of basis $B = \{u_1, u_2\}$ and $B' = \{v_1, v_2, v_3\}$,

$$\text{where } u_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, v_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

- Find matrix of transformation $[T]$ relative to the standard basis of R^2 .
- Find matrix of transformation $[T]_{B', B}$ relative to the basis B and B' .

Question#7: [10] Suppose $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ & define the column vectors as $u_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $u_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$,

$u_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Use Gram-Schmidt process to find orthogonal set of vectors $\{v_1, v_2, v_3\}$ and then find

Orthonormal set of vectors $\{q_1, q_2, q_3\}$ by considering standard Euclidean Inner Product between vectors.

Question # 8: [10] Define Orthogonal, Hermitian and Unitary matrix, check whether given matrix A is Orthogonal, Hermitian or Unitary,

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$